Tomohiko Nakamura

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Research Interests

Signal-processing-inspired deep learning, audio and music signal processing, and machine learning

Job

Senior Researcher **Apr. 2023–Present**
National Institute of Advanced Industrial Science and Technology (AIST), Japan.

Visiting Researcher **Sept. 2025–Present**
Graduate School of Information Science and Technology, The University of Tokyo, Japan.

Project Research Associate **Sept. 2019–Mar. 2023**
Graduate School of Information Science and Technology, The University of Tokyo, Japan.

Researcher **Apr. 2016–Aug. 2019**
Intelligent Systems Laboratory, SECOM, Japan.

Research Fellow (DC2) **Apr. 2015–Mar. 2016**
Japan Society for the Promotion of Science (JSPS), Japan.

Education

Ph.D. degree in Information Science and Technology **Mar. 2016**
Graduate School of Information Science and Technology, The University of Tokyo, Japan.

Master's degree in Information Science and Technology **Mar. 2013**
Graduate School of Information Science and Technology, The University of Tokyo, Japan.

Bachelor's degree in Engineering **Mar. 2011**
Faculty of Engineering, The University of Tokyo, Japan.

Teaching

Applied Gaussian Process and Machine Learning **Dec. 2021**
Graduate School of Information Science and Technology, The University of Tokyo, Japan.

Advanced Signal Processing **June 2020, 2022, 2024, and 2026**
Graduate School of Information Science and Technology, The University of Tokyo, Japan.

Student Experiment **Apr. 2020–Mar. 2023**
Department of Mathematical engineering and information physics, The University of Tokyo, Japan.

Languages

Japanese (native), English (basic)

Competitive Funds

Variable time-resolution audio processing framework adaptive to acoustic scenes
JSPS PRESTO (Fundamental Innovation for Real-World Intelligent Systems) *Apr. 2023–Mar. 2027*

Development of deep-layered analysis-by-synthesis techniques for acoustic scene analysis with human intervention
JSPS KAKENHI *Apr. 2023–Mar. 2027*

Sampling-frequency-independent deep learning for audio media processing

JST ACT-X (Frontier of Mathematics and Information Science) *Oct. 2021–Mar. 2024*

+ 4 funds as representative, 5 funds as co-researcher, and 1 fund as research participants.

Publications

Journal Papers

[1] Yoshiaki Bando, Tomohiko Nakamura, Satoru Fukayama, and Shinji Watanabe, "Online frontend system for multi-talker DSR using neural blind source separation and diarization," *IEEE Transactions on Audio, Speech and Language Processing*, July 2026.

[2] Ren Uchida, Kohei Yatabe, and Tomohiko Nakamura, "Encoder-masking-decoder networks using orthogonal convolutional layer as invertible linear encoder," *Acoustical Science and Technology*, vol. advpub, no. e26.10, May 2026.

[3] Kanami Imamura, Tomohiko Nakamura, Norihiro Takamune, Kohei Yatabe, and Hiroshi Saruwatari, "Stride conversion algorithms for convolutional layers and its application to sampling-frequency-independent deep neural networks," *Signal Processing*, vol. 242, Nov. 2025.

[4] Yuki Ito, Tomohiko Nakamura, Shoichi Koyama, Shuichi Sakamoto, and Hiroshi Saruwatari, "Spatial upsampling of head-related transfer function using neural network conditioned on source position and frequency," *IEEE Open Journal of Signal Processing*, vol. 6, pp. 1109–1123, Sept. 2025.

[5] +11 papers

Peer-Reviewed International Conferences and Workshops

[1] Tomohiko Nakamura, Wataru Nakata, Kanami Imamura, and Yuki Saito, "Neural audio codec with adjustable token temporal resolution using sampling-frequency-independent convolutional layers," in *Proceedings of International Workshop on Acoustic Signal Enhancement*, Sept. 2026.

[2] Ege Erdem, Shoichi Koyama, Tomohiko Nakamura, Orchisama Das, and Zoran Cvetkovic, "SF-Flow: Sound field magnitude estimation via flow matching guided by sparse measurements," in *Proceedings of International Workshop on Acoustic Signal Enhancement*, Sept. 2026.

[3] Yuto Ishikawa, Norihiro Takamune, Kouei Yamaoka, Tomohiko Nakamura, and Hiroshi Saruwatari, "Joint optimization of demixing filters and asymmetric window function for independent vector analysis," in *Proceedings of International Workshop on Acoustic Signal Enhancement*, Sept. 2026.

[4] +47 papers

Invited Talks

[1] Tomohiko Nakamura, "Trends and prospects for audio source separation using deep learning," *Meeting on Technical Committee on Engineering Acoustics, IEICE*, Mar. 2025. (in Japanese)

[2] Daichi Kitamura, Tomohiko Nakamura, "Fundamentals and applications of audio source separation — A guide to becoming an expert," *2023 Otogaku Symposium*, Jun. 2023. (in Japanese)

[3] Tomohiko Nakamura, "Signal-processing-inspired deep learning," *IEEE NZ Signal Processing/Information Theory Joint Chapter in co-hosted by the Acoustics Research Centre, University of Auckland*, Dec. 2022.

[4] +1 invited presentation

Overview Papers

[1] Shoichi Koyama, Juliano Ribeiro, Tomohiko Nakamura, Natsuki Ueno, and Mirco Pezzoli, "Physics-informed machine learning for sound field estimation: Fundamentals, state of the art, and challenges," *Special Issue on Model-Based and Data-Driven Audio Signal Processing, IEEE Signal Processing Magazine*, vol. 41, pp. 60–71, 2024.

[2] +1 paper

Patents

[1] Tomohiko Nakamura, "Object recognition device, method, and program," Japan Patent JP7349288, 13-Sept-2023.

[2] +10 patents

Awards

1. The Awaya Kiyoshi Research Award, ASJ, Mar. 2024.
2. The Itakura Prize Innovative Young Researcher Award, ASJ, Mar. 2022.
3. +12 awards